Installing gclip into an Existing Paperclip Instance

This guide walks you through installing gclip's 9 agents and 37 skills into a running Paperclip instance that already has an organization in place.

Prerequisites

- A running Paperclip instance with board access
- An existing company with agents already deployed
- Git installed locally

```
git clone https://github.com/boosey/gclip.git
cd gclip
```

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1. Understand the Mapping

gclip defines 9 engineering-focused agents. Your existing org likely has some overlap. Here's how to think about the mapping:

Your Current Org

```
CEO
├── CIO
│   └── CTO
│       ├── Senior Developer & Tech Lead
│       │   └── Frontend Engineer
│       └── UX/UI Designer
├── General Counsel (Legal Research)
├── CFO
├── COO
└── CMO
```

gclip Agents → Your Org

gclip Agent	Action	Maps To	Notes
CEO	ENHANCE	Your existing CEO	Add `/office-hours` for business strategy framing; delegates dev process to CIO
CIO	ENHANCE	Your existing CIO	Major upgrade — becomes the dev process owner with autoplan, scope challenge, workflow gates
CTO	ENHANCE	Your existing CTO	Add gclip skills + cognitive config to existing agent
Staff Engineer	ENHANCE	Your existing Senior Developer	Rename optional — add review/investigate/guard skills
UX Designer	ENHANCE	Your existing UX/UI Designer	Add full gclip design skill suite (11 skills)
Release Engineer	CREATE	New agent under CTO	No existing equivalent — handles ship/deploy pipeline
QA Engineer	CREATE	New agent under CTO	No existing equivalent — handles testing/browser automation
CSO	CREATE	New agent under CTO	No existing equivalent — handles security audits
Evaluator	CREATE	New agent under CEO	Novel — the self-improvement engine
Learning Curator	CREATE	New agent under Evaluator	Novel — institutional memory manager

Agents You Keep As-Is (Outside gclip Scope)

These agents are not part of gclip's engineering focus. They stay untouched:

- **General Counsel** — Legal research. No gclip equivalent.
- **CFO** — Finance. No gclip equivalent.
- **COO** — Operations. No gclip equivalent.
- **CMO** — Marketing. No gclip equivalent.
- **Frontend Engineer** — Reports to Senior Developer. gclip doesn't have a separate frontend role — your Senior Developer (Staff Engineer equivalent) delegates to them. Keep as-is.

Your Resulting Org After Installation

```
CEO
├── CIO
│   └── CTO (enhanced with gclip skills)
│       ├── Senior Developer (enhanced → Staff Engineer skills)
│       │   └── Frontend Engineer (unchanged)
│       ├── UX/UI Designer (enhanced with 11 gclip skills)
│       ├── Release Engineer (NEW)
│       ├── QA Engineer (NEW)
│       └── CSO (NEW)
├── General Counsel (unchanged)
├── CFO (unchanged)
├── COO (unchanged)
├── CMO (unchanged)
├── Evaluator (NEW)
│   └── Learning Curator (NEW)
```

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2. Enhance Existing Agents

For each existing agent that maps to a gclip role, you need to update their system prompt (AGENTS.md content) and add skills.

2a. Enhance Your CEO

Your CEO remains a business leader. The gclip enhancement is minimal — the CEO gets `/office-hours` for company-level strategy sessions and learns to delegate the dev process entirely to the CIO.

Open your CEO agent in the Paperclip UI (Settings → Agent Configuration).

Add skills: `office-hours`

Update system prompt — replace with content from `agents/ceo/AGENTS.md`. Key changes: the CEO explicitly delegates the development process to the CIO and intervenes only on cross-functional conflicts, budget thresholds, and Evaluator proposals with strategic impact.

2b. Enhance Your CIO (THE BIG ONE)

This is the most important enhancement. Your CIO becomes the **owner and enforcer of the gclip development methodology**. In gstack's original design, the "CEO" role was really a dev process orchestrator — in your org, that's the CIO.

Add skills: `autoplan`, `office-hours`, `plan-ceo-review`, `workflow-gate`, `learn`

Replace system prompt — use the full content from `agents/cio/AGENTS.md`. The CIO now:

- **Owns Think → Plan → Build → Review → Test → Ship → Reflect** and enforces phase ordering
- **Runs scope challenge** via `/plan-ceo-review` on every engineering initiative (EXPANSION / SELECTIVE_EXPANSION / HOLD / REDUCTION, with bias toward REDUCTION)
- **Orchestrates autoplan** — CIO → Design → Eng review pipeline (sequential, gated)
- **Enforces workflow gates** — the `/workflow-gate` skill blocks agents from acting on issues in wrong phases
- **Translates CEO directives** into engineering scope — finds the smallest scope that achieves the business objective

Set cognitive config:

```
cognitive:
  gearing: user-sovereignty
  decision_default: TASTE
  workflow_phases: [think, plan]
  anti_sycophancy: true
```

Note: The CIO keeps `/office-hours` for engineering-scoped problem framing. The CEO also has `/office-hours` for company-level strategy. Same skill, different context.

2c. Enhance Your CTO

Add skills: `plan-eng-review`, `plan-design-review`, `retro`, `codex`

Update system prompt — append content from `agents/cto/AGENTS.md`. Key additions:

- Search-first cognitive gearing
- Engineering and design plan review capabilities
- Weekly retro production
- Issue status transition authority (todo → in_progress)

Note: Your CTO reports to CIO, which is exactly how gclip is now designed. The CIO orchestrates the overall process; the CTO executes the technical reviews within it.

2d. Enhance Your Senior Developer → Staff Engineer

Add skills: `review`, `investigate`, `guard`, `learn`

Update system prompt — append content from `agents/staff-engineer/AGENTS.md`. Key additions:

- Boil-the-lake cognitive gearing
- Two-pass adversarial code review

- Root-cause investigation with 3-strike rule
- Destructive command guard
- Operational learning logging

Operational limits to configure:

- `max_fixes_per_heartbeat: 50`
- `strike_rule: 3`
- `wtf_likelihood_threshold: 0.7`

Optional: Rename from "Senior Developer" to "Staff Engineer" for consistency with gclip terminology.

2e. Enhance Your UX/UI Designer

Add skills: `design-consultation`, `design-review`, `design-html`, `design-shotgun`, `user-research`, `wireframe`, `information-architecture`, `interaction-design`, `ab-test-design`, `browse`, `learn`

Update system prompt — append content from `agents/ux-designer/AGENTS.md`. This is the biggest enhancement — your UX/UI Designer goes from a general design role to a full UX lifecycle agent covering research, wireframing, IA, design systems, visual auditing, implementation, interaction design, and experimentation.

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3. Create New Agents

For each new agent, create them through the Paperclip UI or API.

3a. Create Release Engineer

Via Paperclip UI: Agents → Create Agent

Field	Value
Name	Release Engineer
Title	Release Engineer
Reports To	CTO
Adapter	claude_local
Budget	$300/mo (30000 cents)
Context Mode	thin

Field	Value
Heartbeat	Event-triggered

Skills: `ship`, `land-and-deploy`, `document-release`, `setup-deploy`

System prompt: Copy the full markdown body from `agents/release-engineer/AGENTS.md`

3b. Create QA Engineer

Field	Value
Name	QA Engineer
Title	QA Engineer
Reports To	CTO
Adapter	claude_local
Budget	$400/mo (40000 cents)
Context Mode	fat
Heartbeat	Event-triggered

Skills: `browse`, `qa`, `qa-only`, `benchmark`, `canary`, `setup-browser-cookies`

System prompt: Copy from `agents/qa-engineer/AGENTS.md`

3c. Create CSO (Chief Security Officer)

Field	Value
Name	CSO
Title	Chief Security Officer
Reports To	CTO
Adapter	claude_local
Budget	$300/mo (30000 cents)
Context Mode	fat
Heartbeat	Event-triggered

Skills: `cso-audit`, `investigate`, `careful`, `freeze`, `unfreeze`, `guard`

System prompt: Copy from `agents/cso/AGENTS.md`

3d. Create Evaluator

Field	Value
Name	Evaluator
Title	Agent Performance Evaluator
Reports To	CEO
Adapter	claude_local
Budget	$600/mo (60000 cents)
Context Mode	fat
Heartbeat	Weekly (Monday 02:00 UTC) + event (on issue close)

Skills: `evaluate`, `propose-improvement`, `retro`, `learn`

System prompt: Copy from `agents/evaluator/AGENTS.md`

Important: The Evaluator reports directly to the CEO, not to the CIO/CTO chain. This is intentional — it observes all agents independently and cannot be influenced by the engineering hierarchy it evaluates.

3e. Create Learning Curator

Field	Value
Name	Learning Curator
Title	Institutional Memory Manager
Reports To	Evaluator
Adapter	claude_local
Budget	$200/mo (20000 cents)
Context Mode	thin
Heartbeat	Weekly (Monday 06:00 UTC)

Skills: `learn`, `retro`

System prompt: Copy from `agents/learning-curator/AGENTS.md`

Environment variables: Set `GH_TOKEN` (optional) if you want the Learning Curator to create PRs for approved agent definition changes.

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4. Install Skills

Each gclip skill has a `SKILL.md` file in `skills/{skill-name}/SKILL.md`. Skills need to be registered in your Paperclip instance so agents can invoke them.

How to Install a Skill

For each skill directory in `skills/`:

1. Open Paperclip UI → Skills → Create Skill
2. **Name:** Use the directory name (e.g., `review`, `qa`, `cso-audit`)
3. **Description:** Copy the `description` field from the SKILL.md frontmatter
4. **System prompt:** Copy the full markdown body from the SKILL.md file
5. **Assign to agents:** Check the agent table below

Skill → Agent Assignment Reference

Strategic (CEO + CIO)

- `office-hours` → CEO (business strategy), CIO (engineering initiatives)
- `autoplan` → CIO
- `plan-ceo-review` → CIO

Engineering (CTO)

- `plan-eng-review` → CTO
- `plan-design-review` → CTO
- `codex` → CTO

Code Quality (Staff Engineer / Senior Developer)

- `review` → Staff Engineer
- `investigate` → Staff Engineer, CSO
- `guard` → Staff Engineer, CSO

Design (UX/UI Designer)

- `design-consultation` → UX Designer

- `design-review` → UX Designer
- `design-html` → UX Designer
- `design-shotgun` → UX Designer
- `user-research` → UX Designer
- `wireframe` → UX Designer
- `information-architecture` → UX Designer
- `interaction-design` → UX Designer
- `ab-test-design` → UX Designer

Testing (QA Engineer)

- `qa` → QA Engineer
- `qa-only` → QA Engineer
- `benchmark` → QA Engineer
- `canary` → QA Engineer
- `browse` → QA Engineer, UX Designer
- `setup-browser-cookies` → QA Engineer

Security (CSO)

- `cso-audit` → CSO
- `careful` → CSO
- `freeze` → CSO
- `unfreeze` → CSO

Shipping (Release Engineer)

- `ship` → Release Engineer
- `land-and-deploy` → Release Engineer
- `document-release` → Release Engineer
- `setup-deploy` → Release Engineer

Workflow (CIO)

- `workflow-gate` → CIO

Self-Improvement (Evaluator + Learning Curator)

- `evaluate` → Evaluator
- `propose-improvement` → Evaluator
- `retro` → CTO, Evaluator, Learning Curator
- `learn` → CIO, Staff Engineer, UX Designer, Evaluator, Learning Curator

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5. Configure the Self-Improvement Loop

The self-improvement loop requires scheduled heartbeats for three agents:

Heartbeat Schedule

Agent	Schedule	Purpose
CTO	Sunday 23:00 UTC	Weekly retro (metrics gathering)
Evaluator	Monday 02:00 UTC	Performance evaluation + proposal generation
Learning Curator	Monday 06:00 UTC	Apply approved changes + prune learnings

Set these in Paperclip UI → Agent → Heartbeat Configuration.

The 3-hour gaps between heartbeats ensure each agent's output is available as input for the next.

Board Approval Workflow

When the Evaluator generates improvement proposals, they appear in:

Paperclip UI → Board → Pending Approvals

Each proposal includes:

- **Summary** — what change is proposed
- **Evidence** — metrics and data points supporting the proposal
- **Proposed change** — specific diff to agent definitions or configuration
- **Rollback plan** — how to undo if the change doesn't work

You (the board) approve or reject each proposal. Approved changes are applied by the Learning Curator on its next heartbeat.

Initialize the Learning Store

Create the institutional memory file in your Paperclip workspace:

```
# In your Paperclip company workspace
mkdir -p .context/retros
echo '[]' > .context/learnings.json
```

This gives agents a place to write and read operational learnings.

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6. Configure the Ethos

Copy `ETHOS.md` into your Paperclip company's shared documents. This is the cognitive philosophy all gclip agents reference. There are two ways to make it available:

Option A — Shared document (recommended): Upload `ETHOS.md` as a company-wide document in Paperclip UI → Documents → Upload. All agents with `fat` context mode will receive it automatically.

Option B — Prepend to each agent prompt: Add a reference line to each enhanced/new agent's system prompt:

```
You operate under the gclip ethos. See ETHOS.md for the full cognitive philosophy.
```

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7. Verify the Installation

After all agents are created and skills installed, verify with this checklist:

Agent Verification

- [ ] CEO has 1 gclip skill (office-hours)
- [ ] CIO has 5 gclip skills (autoplan, office-hours, plan-ceo-review, workflow-gate, learn)
- [ ] CTO has 4 gclip skills (plan-eng-review, plan-design-review, retro, codex)
- [ ] Senior Developer has 4 gclip skills (review, investigate, guard, learn)
- [ ] UX/UI Designer has 11 gclip skills (all design + browse + learn)
- [ ] Release Engineer exists with 4 skills (ship, land-and-deploy, document-release, setup-deploy)
- [ ] QA Engineer exists with 6 skills (browse, qa, qa-only, benchmark, canary, setup-browser-cookies)
- [ ] CSO exists with 6 skills (cso-audit, investigate, careful, freeze, unfreeze, guard)
- [ ] Evaluator exists with 4 skills (evaluate, propose-improvement, retro, learn)
- [ ] Learning Curator exists with 2 skills (learn, retro)

Hierarchy Verification

- [ ] Release Engineer, QA Engineer, CSO report to CTO
- [ ] Evaluator reports to CEO (not CIO or CTO)
- [ ] Learning Curator reports to Evaluator
- [ ] CIO is enhanced with dev process skills (not unchanged)
- [ ] Existing agents (General Counsel, CFO, COO, CMO, Frontend Engineer) are unchanged

Heartbeat Verification

- [ ] CTO has weekly heartbeat (Sunday 23:00 UTC)
- [ ] Evaluator has weekly heartbeat (Monday 02:00 UTC)
- [ ] Learning Curator has weekly heartbeat (Monday 06:00 UTC)
- [ ] All other new agents are event-triggered

Self-Improvement Verification

- [ ] `.context/learnings.json` exists and is readable by all agents
- [ ] `.context/retros/` directory exists
- [ ] ETHOS.md is accessible as a shared document
- [ ] Board approval queue is accessible at Board → Pending Approvals

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8. Recommended Rollout Order

Don't install everything at once. Roll out in phases:

Week 1: Core Engineering

1. Enhance CTO with gclip skills
2. Enhance Senior Developer with Staff Engineer skills
3. Create QA Engineer
4. Create Release Engineer

This gives you the basic Think → Plan → Build → Review → Test → Ship pipeline.

Week 2: Design + Security

5. Enhance UX/UI Designer with full skill suite
6. Create CSO

This adds design rigor and security auditing to the pipeline.

Week 3: Self-Improvement

7. Enhance CEO with gclip skills
8. Create Evaluator
9. Create Learning Curator
10. Configure heartbeat schedules

11. Initialize learning store

This activates the self-improvement loop. Give it 2-3 weeks of data before expecting meaningful proposals from the Evaluator.

Troubleshooting

Agent can't find skills

Skills must be registered in Paperclip before agents can invoke them. Check Skills → list to verify all 37 skills are registered.

Evaluator has no data

The Evaluator needs heartbeat history, cost events, and closed issues to analyze. It won't produce useful proposals until agents have been running for at least 1-2 weeks.

Workflow gate blocks everything

The workflow-gate skill enforces phase ordering. If an agent is blocked, check the issue's current status — it may be in the wrong phase for that agent's role. Fix by updating the issue status in Paperclip.

Budget alerts

Default budgets in gclip are conservative. If agents are hitting 80% warnings frequently, the Evaluator will eventually propose budget adjustments. Or adjust manually in Agent → Settings → Budget.

Learning store permissions

All agents with the `learn` skill need read/write access to `.context/learnings.json`. Ensure the Paperclip workspace has appropriate file permissions.